

1. What kind of software & hardware are/will be on our drone?

- Hardware

- Wifi Antenna
- GPS receiver (2) / other sensors
- Cube (flight controller)
- 360 auto-stabilize gimbal camera
- NVIDIA Jetson Orin Nano (onboard computer)

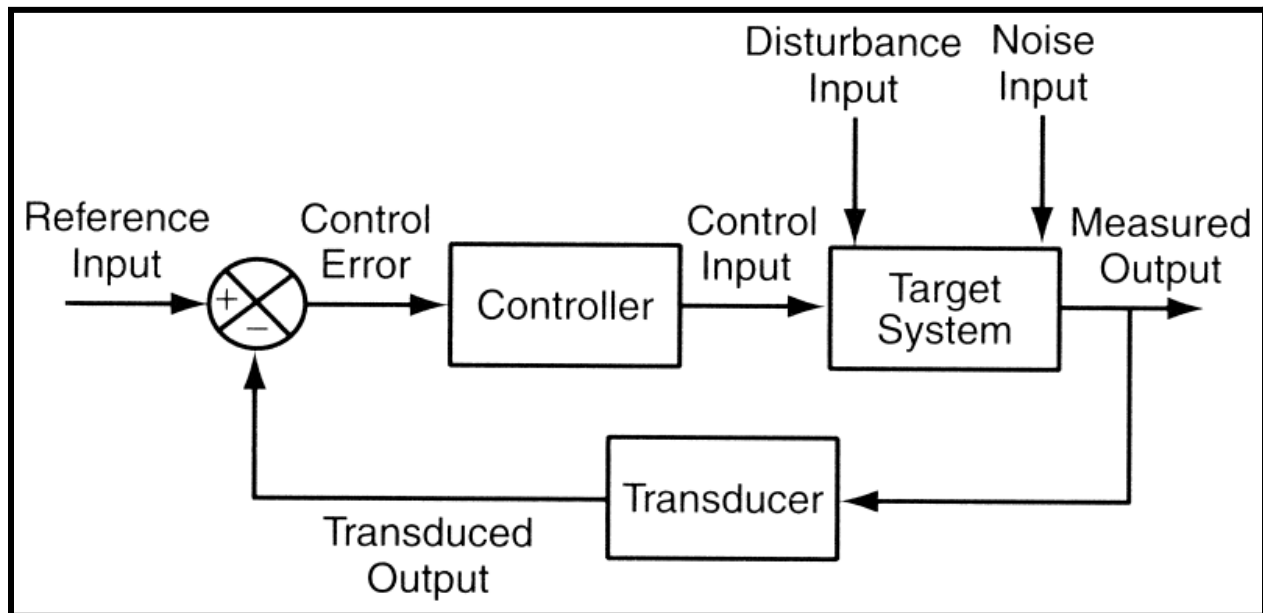
- Software

- Object detection (**Week 3**)
 - Historically done with OpenCV (traditional computer vision library)
 - Last year, we switched to YOLO (manually supervised learning)
- Aerial Imagery/Stitching (**Week 2**)
 - A whole bunch of theory surrounding it, people have been perfecting this for 15+ years
 - We tried OpenCV but had better results with OpenDroneMap so far
 - Currently, we do this offline

Direct Comparison

Feature	OpenCV	YOLO
Core Nature	A comprehensive software library.	A single deep learning network architecture.
Primary Focus	General image processing & pixel manipulation.	Real-time multi-object detection & tracking.
Underlying Tech	Traditional math/matrix algorithms + basic AI.	Convolutional Neural Networks (CNNs).
Hardware Needs	Lightweight; runs efficiently on standard CPUs.	Demanding; requires GPUs for fast training/inference.
Scope of Tasks	Broad (filtering, resizing, color tracking, camera IO).	Narrow (predicts bounding boxes and class labels).

- Flight controller (Week 4)
 - There are two parts to an FC
 - Hardware (e.g. Cube Orange+, Pixhawk)
 - Firmware (e.g. ArduPilot, Betaflight, PX4)
 - All open source
 - It keeps the drone stable and flying
 - Has very little autonomy
 - Takes commands via MAVLink over a physical link (e.g. UART serial cable) from onboard computer (e.g. move North at 10 m/s)



- Feedback controllers tells drone:
 - Move 5 meters
 - Checks to see if they actually made 5 meters
 - Makes sure it doesn't crash
- Onboard computer, sensors & middleware (Week 4, 5 & 6)
 - Jetson/Raspberry Pi running Ubuntu; main “brain” of system
 - Flight controllers are not intelligent, need to receive commands from somewhere (in this case Ubuntu)
 - It runs code (written under ROS framework) that tells the FC what to do
 - Typically, the code takes FC and/or (filtered) sensor data (position, velocity, body rates, etc) as input, processes these data using the logics/algorithms you will implement, and generates fly commands. ROS will help you get data from and send commands to the hardware :)
- ROS will connect firmware and software

2. What kinds of challenges/projects can YOU work on as a Software team member?

- Projects are done for two main competitions
 - SUAS in next September
 - CUASC in May
- Typical Challenges:
 - Time-based waypoint flying
 - Mapping/surveying
 - Object detection, classification & localization
 - Depending on your approach, mapping can fit under this
 - Package drop/payload delivery at target(s)
- Individual/Group Projects
 - GPS-denied navigation, mapping and/or localization
 - (Learning-based) controller research (how to make the drone fly fast)
 - Swarming

3. What's the state of modern drone autonomy software?

- Basic Autonomy vs Production Autonomy

Feature	Basic Stack (good starting point)	Production Stack (what we're working towards)
Flight control	ArduPilot/PX4/Betaflight	PX4 or ArduPilot on FC (with necessary customization)
Higher-level software	Python scripts, simple onboard computer apps, or none	ROS 2 or custom middleware on companion computer
Autonomy	Waypoint following	Perception + object localization + mission logic
Robustness & safety	Basic failsafes	Redundant sensing, health monitoring, extensive logging, etc

- BVLOS is growing in the industry
- BVLOS: Beyond Visual Line of Sight
 - Also multiple drones by one operator working together (swarming)!

4. How does one learn about drones effectively in 2026?

- AI is certainly a huge help
 - Great for conceptual questions
 - e.g. “what is SLAM” or “How do drones stay balanced”
 - → SLAM stands for simultaneous localization and mapping
 - Great for help with code if used correctly and even better with prior human expertise
 - Great for explaining papers!
- EXPERIENCE EXPERIENCE EXPERIENCE
- Sample Workflow:
 - Talk back and forth with AI to make sure you understand it at a broad level
 - SET GOALS FOR YOURSELF. DO PROJECTS.
 - Find ways of tracking your progress so you stay encouraged and feel like you’re doing something
 - For individual details, a combination of AI and fact-checking on individual sites/personal experience is best. AI isn’t perfect with specifics
 - Reach out to your friends if you need help